Previous-year hourly electricity forecasts the next year

Abstract

We forecast hourly electricity in the next calendar year from the same building's previous year. On site Rat of the Building Data Genome Project 2, the hour-of-week mean is fit only on hours before 2017-01-01T00:00:00. 8 other buildings set a split-conformal interval and 8 buildings are scored on the later year. The interval covered 65865/69506 test-year hours (0.9476) at a nominal coverage of 0.9000. Its mean width was 49.8700 kWh. The mean test-year load was 36.2297 kWh and the mean absolute error was 9.9583 kWh. A profile transferred from other buildings, fit without the target building's own year, had mean absolute error 14.7358 kWh and mean width 151.8214 kWh on the same hours.

Keywords

building electricity; next-year forecast; prediction interval; Building Data Genome 2

1. Question

The question is whether the next year of hourly electricity on a building can be forecast from that building's previous year, with an interval that is calibrated on other buildings and narrow enough to sit near the load.

2. Data

The meters are the cleaned hourly electricity file and the building metadata in Building Data Genome Project 2, release v1.0 (doi:10.5281/zenodo.3887306), described by Miller et al. (2020). The data license is CC BY 4.0. The ASHRAE Kaggle competition files were not used. A meter is kept when at least 95 percent of its hours are present and its floor area is positive. The site with the most such meters is used. At most 40 meters are kept, spread across the sorted building identifiers. Each kept building is assigned to training, calibration, or test by a fixed hash of its identifier. Hours before 2017-01-01T00:00:00 are the previous year. Later hours are the test year. The point forecast for a building uses only that building's previous year. The interval quantile uses the test year of the calibration buildings and does not use the test buildings. The training buildings are used only for the transferred-profile comparison.

3. Forecast

The point forecast is the mean load at that hour of the week in the previous year of the same building. The interval is inductive split-conformal regression with a locally weighted absolute residual (Vovk, Gammerman and Shafer, 2005; Lei et al., 2018). The weight is that building's mean absolute residual in the previous year, which is known before the test year. The quantile uses rank ceil((n+1)*(1-alpha)) on the calibration buildings' test-year scores, with alpha equal to 0.10. The half-width on a building is that quantile times its previous-year mean absolute residual. The comparison forecast is a usage-specific hour-of-week mean of kWh per square metre, fit on the previous year of the training buildings and multiplied by floor area, with its own area-weighted split-conformal interval. A language model assisted with wording. It is not an author. The measurements were checked against the script output.

4. Measurement

Site Rat: 24 train buildings, 8 calibration buildings, 8 test buildings. Test year starts at 2017-01-01T00:00:00.

Own-history forecast on the test year: 65865/69506 hours inside the interval (0.9476), nominal coverage 0.9000, mean width 49.8700 kWh.

Own-history MAE 9.9583 kWh. Mean test-year load 36.2297 kWh.

Profile transferred from other buildings' previous year, same test hours: 65880/69506 (0.9478), mean width 151.8214 kWh, MAE 14.7358 kWh.

Per test building, own-history coverage min 0.8593, median 0.9658, max 0.9925.

5. Reading the measurement

On these held-out buildings the previous-year profile has a lower mean absolute error than the profile transferred from other buildings. The measured coverage is at or above the nominal level. The mean interval width is less than twice the mean test-year load. The nominal level applies to the pooled test-year hours. The lowest building in the measurement section can fall below that level. A symmetric interval can extend below zero. Weather is not an input. One site is not another site. This is not a building-management deployment.

6. Conclusion

On these held-out buildings the previous-year profile has a lower mean absolute error than the profile transferred from other buildings. The measured coverage is at or above the nominal level. The mean interval width is less than twice the mean test-year load. The file reports that forecast and no stronger one.

References

Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. Distribution-free predictive inference for regression. Journal of the American Statistical Association, 113(523), 1094-1111, 2018.

Miller, C., et al. The Building Data Genome Project 2. Scientific Data, 7, 368, 2020. doi:10.1038/s41597-020-00712-x.

Vovk, V., Gammerman, A., and Shafer, G. Algorithmic Learning in a Random World. Springer, 2005.

Dataset archive: doi:10.5281/zenodo.3887306. License: CC BY 4.0.